Lecture 1. Combinatorial Analysis

Counting principle, permutations, combinations, binomial and multinomial theorems, integer solutions

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Probabilities on finite spaces with equally likely outcomes reduce to counting.

1.1 The basic principle of counting

Motivation. Equally likely probabilities are ratios of counts, and counting reduces to multiplying the choices at each stage.

Theorem 1.1 (Generalized counting principle)

If r experiments are performed in sequence so that the first has n_1 possible outcomes and, for each outcome of the first $i-1$ experiments, the i-th has n_i possible outcomes, then the r experiments together have $n_1 n_2 \cdots n_r$ possible outcomes.

Proof. For $r = 2$ list the outcomes as pairs (i, j), $1 \le i \le n_1$, $1 \le j \le n_2$: an $n_1 \times n_2$ array. Induct on r, treating the first $r-1$ experiments as one. ■

The number n_i may not depend on earlier outcomes, but the *set* of outcomes may.

Example 1.1 (License plates)

Plates of 3 letters and 4 digits: $26^3 \cdot 10^4 = 175{,}760{,}000$; with no repeats, $26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 \cdot 7 = 78{,}624{,}000$. Likewise there are 2^n functions from an n-set to $\{0, 1\}$.

1.2 Permutations

An ordered arrangement of n distinct objects is a **permutation**; there are $n! = n(n-1)\cdots 1$ of them, with $0! = 1$. The number of ordered selections of r objects out of n is $n(n-1)\cdots(n-r+1) = n!/(n-r)!$.

If the objects fall into r groups of $n_1, \ldots, n_r$ indistinguishable ones, each distinguishable arrangement corresponds to $n_1! \cdots n_r!$ labelled permutations, so there are

$$\frac{n!}{n_1!\, n_2! \cdots n_r!}$$

distinguishable arrangements.

Example 1.2

(a) The letters of PEPPER (three P, two E, one R) give $6!/(3!\,2!\,1!) = 60$ arrangements.
(b) Four mathematics, three chemistry, two history and one language book, same subjects together: $4!\,(4!\,3!\,2!\,1!) = 6912$ arrangements.
(c) 6 men and 4 women ranked without ties: $10! = 3{,}628{,}800$; men and women ranked separately: $6!\,4! = 17{,}280$.

1.3 Combinations

The number of *unordered* groups of r objects chosen from n is

$$\binom{n}{r} = \frac{n(n-1)\cdots(n-r+1)}{r!} = \frac{n!}{(n-r)!\,r!}, \qquad 0 \le r \le n,$$

since each group appears $r!$ times among ordered selections. Set $\binom{n}{r} = 0$ for $r < 0$ or $r > n$; $\binom{n}{r} = \binom{n}{n-r}$.

Proposition 1.2 (Pascal's identity)

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r} \text{ for } 1 \le r \le n.$$

Proof. Fix one object. The r-subsets containing it number $\binom{n-1}{r-1}$; those not containing it number $\binom{n-1}{r}$. ■

Theorem 1.3 (Binomial theorem)

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Proof. $(x_1+y_1)\cdots(x_n+y_n)$ is a sum of 2^n products, one per choice of x_i or y_i from each factor; those with k x's correspond to the $\binom{n}{k}$ k-subsets. Set $x_i = x$, $y_i = y$. ■

$x = y = 1$ gives $\sum_k \binom{n}{k} = 2^n$; $x = -1, y = 1$ gives $\sum_k (-1)^k \binom{n}{k} = 0$ $(n \ge 1)$.

Counting one set in two ways:

Proposition 1.4

$$\text{(a)} \sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}, \qquad \text{(b)} \binom{n+m}{r} = \sum_{k=0}^{r} \binom{n}{k}\binom{m}{r-k} \quad \text{(Vandermonde)}.$$

Proof. (a) Count (committee, chair) pairs from n people: committee of size k then chair, or chair (n ways) then any subset of the rest (2^{n-1}).

(b) Choose r people from n men and m women by first fixing the number k of men. ■

Example 1.3 (Antennas in a row)

n antennas in a line, m indistinguishable defectives. Orderings with no two defectives adjacent: the $n-m$ working ones create $n-m+1$ gaps, each taking at most one defective, so $\binom{n-m+1}{m}$. E.g. $n = 4, m = 2$: DWDW, DWWD, WDWD.

1.4 Multinomial coefficients

The number of ways to divide n distinct objects into r *labelled* groups of sizes $n_1, \ldots, n_r$ $(\sum n_i = n)$ is

$$\binom{n}{n_1, \ldots, n_r} = \binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-\cdots-n_{r-1}}{n_r} = \frac{n!}{n_1!\cdots n_r!},$$

the count of Section 1.2 (assigning group labels = arranging a word with n_i copies of label i).

Theorem 1.5 (Multinomial theorem)

$(x_1 + \cdots + x_r)^n = \sum_{n_1+\cdots+n_r=n} \binom{n}{n_1, \ldots, n_r} x_1^{n_1} \cdots x_r^{n_r}$, the sum over nonnegative integer vectors.

Example 1.4 (Labelled versus unlabelled groups)

Ten children into teams A and B of five: $\binom{10}{5,5} = 252$; unlabelled teams: 126.

Example 1.5 (Knockout tournament)

With $n = 2^m$ players, the first round has $n!/\{2^{n/2}(n/2)!\}$ pairings (arrange in a row, pair neighbours, forget orders), each with $2^{n/2}$ outcomes: $n!/(n/2)!$ outcomes. Round by round, the tournament has

$$\frac{n!}{(n/2)!} \cdot \frac{(n/2)!}{(n/4)!} \cdots \frac{2!}{1!} = n!$$

outcomes.

1.5 Integer solutions of equations

Motivation. Stars and bars turns placing identical objects into boxes into choosing separator positions.

Proposition 1.6 (Stars and bars)

There are $\binom{n-1}{r-1}$ vectors of *positive* integers with $x_1 + \cdots + x_r = n$, and $\binom{n+r-1}{r-1}$ vectors of *nonnegative* integers.

Proof. A positive solution chooses $r-1$ of the $n-1$ gaps between n stars as dividers. For nonnegative ones, $y_i = x_i + 1$ solves $\sum y_i = n + r$ positively. ■

So $\binom{n+r-1}{r-1}$ counts placements of n identical balls in r urns and the terms of $(x_1 + \cdots + x_r)^n$.

Example 1.6 (Investment)

\$20,000 in \$1000 units among 4 options: all invested, $\binom{23}{3} = 1771$; not necessarily all (a fifth "cash" variable), $\binom{24}{4} = 10{,}626$; each option at least one unit, $\binom{19}{3} = 969$.

Example 1.7 (Antennas revisited)

In Example 1.3, let $x_1, \ldots, x_{m+1}$ be the working antennas before, between and after the defectives: $\sum x_i = n - m$, $x_1, x_{m+1} \geq 0$, $x_2, \ldots, x_m \geq 1$. With $y_1 = x_1 + 1$, $y_{m+1} = x_{m+1} + 1$, a positive solution of $\sum y_i = n - m + 2$: again $\binom{n-m+1}{m}$. At least two working antennas between defectives: substitute $y_i = x_i - 1$ in the interior.

1.6 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): counting in two ways, and a proof of the multinomial theorem.

Exercise 1.11 (Classifying subsets by their largest element)

Show that for $1 \le k \le n$,

$$\sum_{i=k}^{n} \binom{i-1}{k-1} = \binom{n}{k}.$$

Solution.

Step 1. Let $\mathcal{A}$ be the k-subsets of $\{1,\dots,n\}$, $|\mathcal{A}| = \binom{n}{k}$. Each A has $\max A \ge k$, so $\mathcal{A}$ is the disjoint union of $\mathcal{A}_i = \{A : \max A = i\}$, $i = k,\dots,n$.

Step 2. $A \in \mathcal{A}_i$ is $\{i\}$ plus an arbitrary $(k-1)$-subset of $\{1,\dots,i-1\}$: $|\mathcal{A}_i| = \binom{i-1}{k-1}$.

Step 3. Sum over i. ■

Exercise 1.12 (Committees with a chair and a secretary)

Show that for $n \ge 1$,

$$\sum_{k=1}^{n} k^2 \binom{n}{k} = 2^{n-2} n(n+1).$$

(The identity $\sum_k k\binom{n}{k} = n2^{n-1}$ is Proposition 1.4(a).)

Solution. Count triples (C, a, b): C a nonempty committee from n people, chair $a \in C$, secretary $b \in C$, $a = b$ allowed.

Step 1. (Committee first.) $\binom{n}{k}$ committees of size k, each with k^2 choices of (a,b): $\sum_{k=1}^{n} k^2\binom{n}{k}$.

Step 2. (Officers first, $a = b$.) n choices, the other $n-1$ people free: $n2^{n-1}$.

Step 3. (Officers first, $a \ne b$.) $n(n-1)$ pairs, the other $n-2$ free: $n(n-1)2^{n-2}$ (0 if $n = 1$).

Step 4. Steps 2 and 3 partition the set of Step 1:

$$\sum_{k=1}^{n} k^2 \binom{n}{k} = n2^{n-1} + n(n-1)2^{n-2} = 2^{n-2}(2n + n^2 - n) = 2^{n-2} n(n+1).$$

■

Exercise 1.16 (Tournaments with ties)

A tournament of n contestants ends in an ordering with ties allowed (groups tied for first, next, and so on). Let $N(n)$ be the number of outcomes, $N(0) = 1$.
(a) Show that $N(n) = \sum_{i=1}^{n} \binom{n}{i} N(n-i)$ for $n \ge 1$.
(b) Show that this is equivalent to $N(n) = \sum_{i=0}^{n-1} \binom{n}{i} N(i)$.
(c) Find $N(3)$ and $N(4)$.

Solution.

(a) Classify by the number $i \in \{1,\dots,n\}$ tied for last. Such an outcome is determined by the set L of last players and an arbitrary outcome of the other $n-i$ (one if $i = n$, matching $N(0) = 1$), and conversely; so there are $\binom{n}{i} N(n-i)$ of them. Sum over i.

(b) Substitute $j = n - i$ and use $\binom{n}{n-j} = \binom{n}{j}$.

(c) From (b), $N(1) = N(0) = 1$, $N(2) = N(0) + 2N(1) = 3$,

$$N(3) = N(0) + 3N(1) + 3N(2) = 1 + 3 + 9 = 13,$$
$$N(4) = N(0) + 4N(1) + 6N(2) + 4N(3) = 1 + 4 + 18 + 52 = 75.$$

■

Exercise 1.19 (The multinomial theorem)

Prove Theorem 1.5.

Solution.

Step 1. Expanding the product of n factors gives the sum over words $w = (i_1, \ldots, i_n) \in \{1, \ldots, r\}^n$ of $x_{i_1} \cdots x_{i_n}$, the word recording the summand taken from each factor.

Step 2. If $n_j = \#\{t : i_t = j\}$, then $x_{i_1} \cdots x_{i_n} = x_1^{n_1} \cdots x_r^{n_r}$; grouping words by $(n_1, \ldots, n_r)$,

$$(x_1 + \cdots + x_r)^n = \sum_{n_1 + \cdots + n_r = n} c(n_1, \ldots, n_r)\, x_1^{n_1} \cdots x_r^{n_r},$$

with $c(n_1, \ldots, n_r)$ the number of words with n_j copies of letter j.

Step 3. By Section 1.2, $c(n_1, \ldots, n_r) = n!/(n_1! \cdots n_r!) = \binom{n}{n_1, \ldots, n_r}$. ■